Charotar University of Science and Technology

Education Campus, Changa 388 421.

M Sc (BIOTECHNOLOGY/ MICROBIOLOGY)
SEMESTER II EXAMINATION
BT707/MI707: BIOPROCESS TECHNOLOGY

Date: 05.05.2011
Time: 10.00 AM to 1.00 PM **Total Marks: 70**

Instructions to the candidates:

1. Provide all necessary information as required in the answer sheet.
2. Attempt all questions.
3. Questions in **Part A** should be answered in the question paper itself. Please **do not write** your candidate ID number on the question paper for Part A.
4. Write answers for Section I and Section II of **Part B** in separate answer sheets.

PART A **Total marks: 20**

Q1. Choose the correct option and put √ mark in front of it:

1. Agar-agar is obtained from
   (a) Algae (b) bacteria (c) fungi (d) all of the above
2. The addition of the structural analogues of the product in the production media is a type of
   (a) Induction strategy (b) Nutritional repression strategy
   (c) Feedback resistant mutation strategy (d) None of the above
3. Microbial Type Culture Collection Centre (MTCC) is located
   (a) Unites States of America (b) United Kingdom (c) Japan (d) India
4. Baiting technique is a type of
   (a) Enrichment technique (b) Isolation technique
   (c) Screening technique (d) None of the above
5. A tower fermentor can have aspect ratio of
   (a) 16:1 (b) 1:1 (c) 2:1 (d) 3:1
6. For maximum biomass formation by a bacterial culture used as a biofertilizer, the dissolved oxygen concentration in the production medium should be
   (a) $< C_{crit}$ (b) equal to C_{crit} (c) $> C_{crit}$ (d) none of the above
7. $C_L = -1/K_La\ \{(dC_L/dt)+xQO_2) + C^*$ is the expression for determination of K_La by
   (a) sulfite oxidation technique (b) static gassing out technique
   (c) dynamic gassing out technique (d) none of the above
8. For a stirred tank reactor with baffles the following relation is true
   (a) $tm \propto 1/N$ (b) $tm \propto N$ (c) $tm = N$ (d) none of the above
9. For a fluid if n<1 then the fluid is known as
   (a) pseudoplastic (b) bingham plastic (c) dilatant (d) cassan body

10. $\tau = \tau_0 + n\gamma$ describes the flow behaviour of a

(a) dilatant fluid (b) Bingham plastic fluid (c) pseudoplastic fluid (d) Newtonian fluid

11. $\mu = \mu_{max}$ when S is greater than about

(a) 5 Ks (b) 50Ks (c) 10 Ks (c) 100 Ks

12. If the K_s values of *Saccharomyces sp., Escherichia coli* and *Aspergillus sp.* is 25 mgl^{-1}, 4.0 mgl^{-1} and 5.0 mgl^{-1} respectively, then the affinity of the organisms for glucose in a fermentation medium will be in the order of

(a) *Saccharomyces > Escherichia > Aspergillus* (b) *Escherichia > Saccharomyces> Aspergillus*

(c) *Escherichia> Aspergillus > Saccharomyces* (d) *Aspergillus> Saccharomyces>Escherichia*

13. Under steady state conditions in a chemostat the rate of change in cell concentration is

(a) $dx/dt = 0$ (b) $dx/dt = \mu x$ (c) $dx/dt = GY$ (d) none of the above

14. In a mixed culture parasitism or predation the effect of one organism on the other is expressed as:

(a) + - (b) + + (c) + 0 (d) 0 -

15. Strain degeneration during large scale continuous culture fermentation is known as

(a) back mutation (b) loss of plasmid (c) contamination from within (d) all of the above

16. The following is **not** an example of spontaneous mixed culture processes

(a) food and beverage fermentations (b) waste treatment (c)metal leaching (d) SCP

17. Foaming is the limitation of the following type of centrifuge used for large scale solid- liquid separation

(a) Chamber (b) disc (c) tubular (d) all of the above

18. The following term **is not** used to describe molecules having a hydrophilic portion and a hydrophobic region:

(a) aliphatic (b) surfactants (c) amphipathic (d) detergents

19. The following **is not** the method of concentration of biological material

(a) Evaporation (b) Liquid- liquid extraction (c) Precipitation (d) none of the above

20. The following **is not** used for supercritical fluid extraction of biological products

(a) CO_2 (b) NH_3 (c) dinitrogen oxide (d) bromine

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Part B **Total Marks: 50**

Section I

Q.1(A) Enlist various methods for determination of K_La in fermentation vessels. Discuss the use of "gassing out" method for determination of K_La. (4)

OR

Q.1(A) Giving schematic diagrams describe the use of various type of impellers for designing stirred tank reactors. (4)

Q.1(B) Write a short note on **any one** of the following: (3)
(i) mixing time (ii) methods used for determination of mixing in bioreactors
(iii) spargers

Q.1(C) Explain **any three** of the following terms in one or two sentences: (3)
(i) baffles (ii) Vpump (iii) circualtion time (iv) flooding of the impeller (v) CSTR

Q.2 (A) Discuss: Kinetics of product formation. (4)

OR

Q.2 (A) What are Newtonian and Non Newtonian fluids. Discuss pseudoplastic rheology. (4)

Q.2 (B) Explain **any four** of the following terms in one or two sentences: (4)

(i) Y_{xs} (ii) Y_{ps} (iii) yield stress (iv) specific growth rate (vi) flow index

Q.2 (C) Write a brief note **any one** of the following: (2)
(i) balanced growth (b) casson body rheology (iii) Ks

Section II

Q.3(A) Differentiate between primary and secondary screening of microorganisms. Explain the significance of secondary screening for industrial applications. (4)

OR

Q.3(A) What do you understand by media optimization? Discuss the importance of media optimization in industrial fermentations. (4)

Q.3 (B) Write short note on **any two** of the following: (6)
(i) Containment levels (ii) Use of steam sterilization at industrial scale
(iii) Improvement of productivity of industrial strains of microorganisms using feed-back inhibition.

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Q.4 (A) Differentiate between batch, fed batch and continuous fermentations and give their advantages and limitations. (4)

OR

Q.4 (A) Differentiate between the following: (4)

(i) steady state and Quasi steady state

(ii) Metabolite productivity in Batch vs Continuous process

Q. 4. (B) Write short note on **any one** of the following: (3)

(i) Diauxic growth (ii) Application of mixed culture in food industries

Q.4. (C) Giving suitable examples discuss patterns of mixed substrate utilization in mixed culture cultivation. (3)

Q.5(A) Enlist the steps involved in down stream processing. Explain solid-liquid separation in detail. (4)

OR

Q.5 (A) Enlist various techniques for cell disruption. Explain the technique which you consider most advantageous to recover intracellular product. (4)

Q.5 (B) Write a short note on **any two** of the following: (4)

(i) Filtration (ii) Evaporation (iii) chromatography

Q.5.(C) List the challenges one has to consider in designing downstream processing. (2)

OR

Q.5. (C) Describe briefly liquid-liquid extraction. (2)